

Mitigating Demonstration Bias via Fairness-Aware Trajectory Editing

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ABSTRACT

Imitation-learned mobility policies are trained to reproduce the trajectories sourced from a real taxi fleet — and those trajectories record the fleet’s expertise along with its demographic service inequity, so a faithful imitator learns both. Existing interventions act at two stages of this pipeline: in-processing penalties constrain the learner while the demonstrations stay biased, so the objective and the training signal pull against each other, while synthetic data generation and extensive data modification can reduce realism, introduce distribution shift, and obscure the source of any fairness improvement. We present FATE, a fairness-oriented data augmentation method that instead *edits* a small, attribution-chosen slice of the real trajectories and upweights the edited demonstrations so the fairness survives training. The editor combines two bounded gradient-guided mechanisms under one differentiable objective: *trim* relocates recorded pickups within over-served areas so their measured over-service falls, and *lift* reroutes drivers’ final seeking minutes into under-served areas, with the effect on local supply included in what the objective optimizes. Every edit is confined to a two-cell radius, with a frozen ST-SiameseNet driver-identity discriminator as a realism guardrail. On taxi trajectories from two cities (Shenzhen; San Francisco), editing improves established fairness measures that the objective never optimizes: demographic parity, disparate impact, and the Theil index. Where demand-only editing improves such measures purely by reducing over-service, the supply channel opened by lift adds statistically significant taxi presence — counted as distinct vehicles — to under-served areas in both cities. Upweighted behavior cloning carries the fairness-metric gains into trained policies, and does so specifically through the edited demonstrations: neither a random-upweighting nor a most-fair selection control reproduces the transfer. A targeted demographic oversampling baseline on Shenzhen captures only part of the data-level gain at the cost of fabricating 10.5% of the corpus, and its untargeted placebo degrades fairness outright. FATE thus improves measured fairness without fabricating data and without obscuring which change produced the improvement.

KEYWORDS

fairness, bias mitigation, imitation learning, behavior cloning, data augmentation, trajectory editing, spatiotemporal data mining, urban mobility

1 INTRODUCTION

Urban mobility policies are increasingly learned by imitation: models are trained to reproduce the trajectories real humans actually drove [36]. But trajectories carry more than behavior. They record the fleet’s service allocation — which neighborhoods drivers sought

Bias is learned from data — fairness can be, too

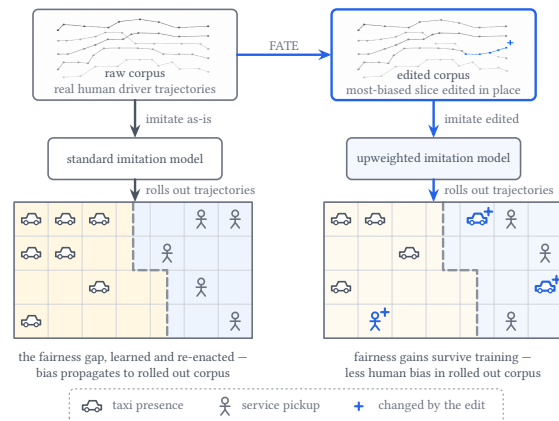


Figure 1: A real fleet’s service demonstrates bias: on Shenzhen taxi GPS trajectories, the advantaged district receives 3.0× the taxi service per unit demand. The trajectories are valuable human expertise, so rather than filter or regenerate them, FATE makes bounded, attribution-chosen edits to the most-biased slice (the *edited corpus*, right). A policy imitating the raw corpus learns the service gap; trained on the upweighted edited corpus, it is measurably fairer. Figure 2 shows the editor.

passengers in, and which they passed over — and in real cities that allocation tracks demographics: housing prices, incomes, migrant population share. Models that imitate such data reproduce the inequity and re-enact it when deployed — a feedback loop documented in other data-driven allocation systems [6, 21].

Interventions typically target one end of the pipeline or the other. In-processing methods regularize the model while leaving the demonstrations biased [37], so objective and training signal conflict. Data-generation and data-rebalancing methods instead alter the demonstrations — synthesizing fair data [31, 34] or reweighting and resampling them [17] — but this can reduce realism, shift the distribution, and obscure the source of any fairness gain. Figure 1 illustrates the gap: the demonstrations are worth preserving, yet a policy that imitates them unchanged learns the service gap they record.

FATE takes a third position. As *data augmentation*, it keeps the real corpus, edits a small slice of its most unfair trajectories, and generates nothing. At most k trajectories are edited ($k = 10,000$ on the primary city, about a tenth of the corpus), each perturbed within a two-cell radius under a frozen driver-identity discriminator (ST-SiameseNet) [28], so realism is inherited from the data rather than re-learned. The edited demonstrations are then *upweighted* during imitation — instance reweighting [17] transplanted

to demonstrations — so the edited slice’s fairness survives training instead of being averaged away.

FATE’s first editing phase, *trim*, moves only demand: it relocates recorded pickups within over-served areas, improving measured fairness on the primary city without helping the under-served group at all. Every one of the pickups it edits originates in an advantaged cell and not one lands in a disadvantaged cell, so the gap closes only by reducing over-service — analogous to the classic leveling-down objection to equalization [25, 27], though here service is relocated rather than removed. This behavior is structural — the constrained optimum of any demand-only editor on this objective (§3.3) — and that diagnosis dictated the second phase. *Lift* reroutes drivers’ final seeking minutes into under-served areas and makes their supply consequences differentiable and endogenous to the objective: the editor is rewarded for providing service where it improves fairness, not merely for redistributing demand. Together, *trim* and *lift* raise demand-adjusted demographic fairness on Shenzhen by +0.0226 and add statistically robust taxi presence to the under-served group (§4.2). On San Francisco the protocol reruns with no algorithm or metric change, and the data-level, external-metric, and downstream conclusions reproduce (+0.0316; §4.6).

Neither edit survives training on its own. The edited slice is a tenth of the corpus, and a vanilla behavior-cloning objective averages it away — a genuine null we verify (§4.3). Upweighting the edited demonstrations recovers the fairness with a monotone dose-response, and two controls pin the recovery to the edit itself: neither upweighting an equal-sized random subset nor upweighting the corpus’s already-fairer trajectories reproduces it. Because a fairness method scored on its own objective proves little, evaluation rides on established measures the objective never optimizes — demographic parity, disparate impact, and the Theil index — computed before and after editing; all three improve on the primary city, every interval excluding zero (§4.2). On the primary city, a targeted demographic-oversampling baseline captures only part of the data-level gain despite fabricating 10.5% of the corpus, and its untargeted placebo degrades fairness outright (§4.4).

Contributions.

- **A fairness-aware trajectory editor.** Two editing phases under one differentiable objective, each driven by an exact attribution mechanism (§3.4).
- **A structural diagnosis of demand-only editing.** Demand-only editing improves fairness without touching the under-served; we show why a differentiable supply channel is the remedy.
- **A transfer recipe with edit-specificity controls.** Upweighted imitation carries fairness into trained policies; controls isolate the edit as the cause.
- **Two-city validation on never-optimized measures.** The full protocol runs on Shenzhen and San Francisco trajectories, scored on external fairness measures the objective never optimizes.

2 RELATED WORK

Fairness interventions in machine learning. Fairness interventions are conventionally grouped by where they act: pre-processing methods repair the training data, in-processing methods constrain

the learner, and post-processing methods adjust its outputs [2]. FATE belongs to the pre-processing family, whose canonical instruments — instance reweighing [17] and disparate-impact repair of feature distributions [7] — were built for labeled classification data. Among formal fairness definitions [32], FATE targets conditional statistical parity [4]: disparities count only after permitted factors are adjusted for. As formulated, none operate on sequential demonstrations or measures fairness over a spatial service distribution — the setting this paper requires. Both instruments reappear here in unfamiliar roles: reweighing applied to edited demonstrations rather than labeled examples, and the parity condition transplanted from a classifier’s predictions to a continuous spatial service ratio as a differentiable editing objective.

Fairness in urban transportation. Transportation equity has its own measurement tradition, in which the Gini index over service distributions is the most widely used instrument [14, 18], alongside entropy- and welfare-based indices [1, 5, 29]. The closest modeling neighbor regularizes a ride-hailing demand predictor at training time, shrinking a racial prediction-error gap from 0.361 to 0.084 [37]. That correction, however, leaves the demonstrations biased: it lives inside one model’s loss and does not transfer to any other consumer of the data. FATE addresses the same bias one stage earlier, on the demonstrations rather than inside the loss, and is scored on equity measures its objective never optimizes.

Imitation learning for mobility, and mobility as identity. Imitation learning is an established route to urban mobility models: conditional GAIL variants recover taxi-driver strategies from trajectories [35, 36], explainable extensions expose the recovered policies [26], and neural simulators generate human mobility [8] — but this literature optimizes realism, and none audits or repairs the demographic fairness of the imitated allocation. A parallel line establishes that individual mobility is highly identifying: trajectory-user linking recovers who generated a trace [10, 24, 38], and Siamese verification decides whether two seeking traces share a driver [28]. FATE contributes no new generator or imitation algorithm; it edits the demonstrations such models consume, and repurposes the identity-verification line as a frozen realism guardrail, avoiding the instability of a live adversarial game [12].

Adversarial perturbation and recourse. The editing machinery descends from adversarial examples: bounded signed-gradient perturbations [11, 19] instantiated spatio-temporally [15], with discrete structure bridged by straight-through estimation [3] and continuous relaxation [16, 22]. Algorithmic recourse showed that the same counterfactual tooling can prescribe beneficial change rather than attack [30, 33], but only to flip one individual’s decision, with no population-level fairness target and no data-realism obligation. FATE extends that constructive reading from the individual to the population: the bounded perturbation edits real trajectories to reduce demographic service inequity, and the ϵ -bound is reinterpreted from an attack-stealth budget to one for preserving driver identity.

Leveling down and feedback loops. Two critiques frame what a fairness gain should mean. The leveling-down objection — equality achieved by making the better-off worse rather than the worse-off better — originates in the ethics of equality [27] and has been applied to machine-learning fairness constraints [25], with data augmentation observed to be the one intervention class that helps the disadvantaged group instead [39]. Feedback-loop analyses show

that allocation systems trained on their own historical output entrench a self-reinforcing bias [6, 21] — in our domain, the continued under-service of already-underserved neighborhoods. Both critiques apply here: our demand-only editor *trim* only manipulates over-service, an analogue of leveling down (§3.3); the supply channel *lift* exists expressly to level up, rerouting real taxi presence into under-served neighborhoods; and because demand recorded where service is thin understates true need, demand endogeneity bounds what any demand-adjusted metric can see.

3 METHODOLOGY

3.1 Problem Formulation

We study taxi service allocation on a city discretized into 0.01° grid cells g and hourly time blocks t ; a pair $i = (g, t)$ with recorded activity is an *active unit*, and N denotes their count ($N \approx 34,500$ on the primary city). Each active unit carries a demand D_i and a supply S_i : demand is its recorded pickups, and supply is active-taxi presence aggregated over the 5×5 cell neighborhood, matching how a cruising taxi is visible to nearby demand. Service received relative to demand is the *service ratio* $Y_i = S_i / \max(D_i, d_0)$, where the demand floor $d_0 = 0.5$ prevents near-empty cells from producing unbounded ratios.

Each cell additionally carries demographic covariates x_i describing its neighborhood: housing price, per-capita compensation, and migrant population share. These covariates resolve at district granularity, which bounds the resolution of any demographic analysis built on them (§3.2).

The data itself is a corpus $\mathcal{T}$ of real per-driver trajectories, each a sequence of passenger-seeking states ending in a pickup. The trajectories are the editable objects; D and S are deterministic aggregations of $\mathcal{T}$, so moving where a driver seeks and picks up moves demand and supply with it.

Task. Given a corpus $\mathcal{T}$ whose service allocation is demographically inequitable, produce an edited corpus $\mathcal{T}'$ that (i) is measurably fairer; (ii) remains realistic at the level of individual trajectories: edited trajectories must remain recognizable to a frozen driver-identity discriminator as their original drivers; and (iii) transfers its fairness into a policy trained on it by imitation. FATE is a *data-augmentation* method: it edits a small, attribution-targeted slice of the real trajectories and generates nothing. At most k trajectories are edited ($k = 10,000$ on the primary city), each perturbed within an L_∞ ball of radius $\varepsilon = 2$ grid cells. The edited slice is then upweighted during downstream training so that its fairness is not averaged away (§3.4).

3.2 The Fairness Objective

FATE edits trajectories by gradient ascent on a single differentiable objective that combines demographic fairness, spatial equity, and realism.

Demand-adjusted demographic fairness (F_{demo} , primary). Raw parity across neighborhoods is the wrong target: taxi service legitimately tracks demand, and a business district *should* see more pickups than a quiet residential block. We therefore measure fairness only in the component of service that demand does not explain (the logic of conditional statistical parity [4], which restricts disparities to those not explained by permitted factors). Stage one

fits a flexible power basis $g_0(D)$ from per-unit demand to the service ratio and takes the residual $R_i = Y_i - g_0(D_i)$; stage two regresses that residual on the demographics and asks how much they explain. We evaluate stage two in closed form rather than through an explicitly fitted regression: with X the z-scored demographic matrix, $\tilde{X} = [1 \ X]$ its design matrix, $H = \tilde{X}(\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top$ the *hat matrix* of that regression (the projection onto the column space of $\tilde{X}$ [13]), and M the centering matrix,

$$F_{\text{demo}} = \frac{R^\top (I - H) R}{R^\top M R} = 1 - r_{\text{demo}}^2. \quad (1)$$

Eq. (1) is the stage-two regression in closed form: the numerator is its residual sum of squares, the denominator its total; H is constant, so each edit step implicitly re-fits the regression exactly. It is smooth, so exact gradients flow through Y to the coordinates being edited, and its projections' idempotence underlies the per-unit attribution of §3.4. The quantity r_{demo}^2 is interpretable as the share of demand-adjusted service inequity attributable to demographics (the Frisch–Waugh–Lovell exactness behind that reading and the $O(N)$ evaluation identity: Appendix A). Boundary cases orient the scale: $R \in \text{span}(X)$ gives $F_{\text{demo}} = 0$ (fully unfair), and $R \perp X$ gives 1 (fully fair).

Two important caveats. First, F_{demo} (*demo* for demographic dependence) is an *associational* quantity: the partial R^2 of a cross-sectional regression on observational district-level demographics, with no identification and no counterfactual. With about ten district profiles the association is ecological, and we make no individual-level claims. Second, adjusting for demand treats recorded demand as exogenous; but demand recorded where service has historically been thin is itself suppressed by that under-service. We return to this in §3.3, where the same assumption turns out to bound what a demand-only editor can achieve.

Spatial equity (F_{spatial} , secondary). F_{spatial} is one minus the mean Gini index of two supply-normalized rates: the *demand-service ratio* $\text{DSR}_i = D_i / S_i$ (pickups per unit of taxi presence) and the *arrival-service ratio* $\text{ASR}_i = A_i / S_i$ (dropoffs per unit). The closed form is in Appendix A. Being demographic-independent, the term acts as a spatial smoothness regularizer complementing F_{demo} .

Realism guardrail (F_{fidelity}). An edit that improves fairness by producing trajectories no real driver would generate is useless for training a realistic imitation model. Because individual mobility signatures are highly identifying [10, 24, 38], we score realism with a frozen driver-identity discriminator of the ST-SiameseNet family [28]: an edited trajectory that still reads as its driver has preserved human fidelity. The discriminator is trained once and never updated during editing. Under FATE's small bounded edits the discriminator's gradient is near zero, so it acts as a realism *guardrail*, not a driver of edits; identity-axis stability under both phases is confirmed empirically and reported as *Fidelity-A* (§4). The distributional realism axis is scored separately at evaluation as *Fidelity-B* (§4.2).

Scalarization. The three terms combine linearly,

$$\mathcal{L} = \alpha_{\text{sp}} F_{\text{spatial}} + \alpha_{\text{de}} F_{\text{demo}} + \alpha_{\text{fi}} F_{\text{fidelity}}, \quad \alpha = (0.1, 0.8, 0.1). \quad (2)$$

The weights are selected *empirically*: a sweep of the weights (fidelity fixed at 0.1), scored across all three metric classes (§4.1), selects (0.1, 0.8, 0.1) by a three-class criterion: maximize ΔF_{demo} subject to $\Delta F_{\text{spatial}} \geq 0$ and the supply-channel lift-up remaining significant under both accounting conventions. The full sweep, with the flatness and monotone-lift-up evidence behind the criterion, is reported in §4.5.

3.3 Why Demand-Only Editing Cannot Help the Under-Served

Relocating pickups is a demand-side lever: on its own it improves the fairness metrics only by redistributing service among better-served areas, while the under-served group gains nothing. On the primary city, every one of the 2,455 pickups edited by the demand-only editor originated in advantaged cells, and not one landed in a disadvantaged cell. The pattern is analogous to the classic leveling-down objection to equalization [27] (measured equality improves while the worst-off gain nothing), recently formalized for machine-learning fairness with a prescription to level *up* instead [25]; data augmentation is the one intervention observed to help the disadvantaged group rather than pull others down [39]. The analogy is inexact in one instructive way: trim *relocates* pickups under conservation rather than removing service, so nothing is destroyed; yet the under-served group is not helped at all.

The behavior is the constrained optimum of the demand-only problem, not an optimizer artifact, for three structural reasons. (i) *Selection*: the deficit attribution that selects trim’s edits (§3.4) concentrates in over-served, high-residual cells, so under-served cells are never edit candidates. (ii) *Leverage*: with supply frozen, $\partial Y_i / \partial D_i = -S_i / D_i^2$ makes padding demand into high-supply advantaged cells more consequential for Y than removing it from starved ones, and 93% of disadvantaged units sit at the demand floor d_0 , where removing demand changes nothing. (iii) *Supply-side inequity*: median taxi presence is 1.8 in disadvantaged versus 17.6 in advantaged cells, and a demand-only editor cannot touch this supply. A greedy unconstrained search, which upper-bounds what any editor could achieve, rules out the last demand-side option: raising the under-served group’s service ratio would require deleting its recorded pickups, which is perverse.

The deeper cause unifies the editor’s limitation with the metric’s. Recorded demand is not exogenous: where service has been thin, demand is suppressed and latent (the feedback-loop pathology documented for other public-allocation systems [6, 21]), so the same censored demand that hides inequity from a demand-adjusted metric leaves a demand-only editor nothing to move in under-served areas.

This sets up the editor’s second phase. The one non-perverse lever is the numerator: $\partial Y_i / \partial S_i = 1 / \max(D_i, d_0) > 0$ everywhere, and at the demand floor, $\Delta Y = 2 \Delta S$, so raising under-served service requires adding taxi *presence*: rerouting real seeking behavior into under-served cells.

3.4 The Editor: Attribution, Trim, and Lift

The objective is optimized one trajectory at a time, so FATE must decide which trajectories are worth editing. Two attribution mechanisms make that decision, one per editing phase: *demand deficit*

attribution locates where unfairness concentrates and drives the demand-side phase, and *supply-gradient attribution* locates where added taxi presence would most improve fairness and drives the supply-side phase.

Both phases repurpose bounded adversarial perturbation [11, 19], in its spatio-temporal instantiation [15], *constructively*, in the spirit of algorithmic recourse [30, 33]: the same counterfactual-perturbation tooling that finds a minimal input change to flip a decision instead prescribes a minimal, bounded, beneficial change. Per iteration, the edit offset takes a signed-gradient step under a cumulative L_∞ clip,

$$\delta \leftarrow \text{clip}(\eta \cdot \text{sign}(\nabla_\delta \mathcal{L}), -\varepsilon, \varepsilon), \quad \delta_{\text{total}} \leftarrow \text{clip}(\delta_{\text{total}} + \delta, -\varepsilon, \varepsilon), \quad (3)$$

so no pickup ever moves more than $\varepsilon = 2$ cells from its recorded location, regardless of iteration count: the ε -bound reinterpreted as an identity-preservation budget rather than an attack-stealth budget. A temperature-annealed soft cell assignment makes the discrete cell counts differentiable in the continuous edit perturbations, so gradients flow from the objective back to the coordinates being edited (Appendix B).

Trim: demand deficit attribution. Because M and $I - H$ are idempotent, the explained share r_{demo}^2 of Eq. (1) admits an exact per-unit partition of the fairness deficit across active units (Appendix A), with a signed variant that separates over- from under-served units. Trim edits the trajectories whose pickups land in the highest-deficit units: each selected pickup relocates within the ε -ball, padding recorded demand into over-served, supply-rich cells so their measured over-service falls (§3.3).

Lift: supply-gradient attribution. The second mechanism asks of every active unit: *how much fairer would the city’s service be with marginally more taxi presence here, at this hour?* The quantity that answers it is the gradient of the objective with respect to supply. To obtain it we evaluate the objective at hypothetically augmented supply $S + \Delta S$ and read $\partial \mathcal{L} / \partial S_i$ by automatic differentiation at $\Delta S = 0$ as the marginal fairness effect of added presence at each of the $N \approx 34,500$ active units, in one backward pass. The result is a *value-of-presence* map over the city: large where added presence would most repair the fairness terms, and small or negative where presence is already abundant. Its F_{demo} component has a closed form, given with its autograd verification in Appendix A.

To turn the map into candidates, a linearized-offset screen ranks trajectories whose tails could bend into high-value cells, scoring each by its best bounded tail translation (Appendix B). A tail is the pickup itself plus up to four seeking states immediately before it; $\approx 80k$ of the corpus’s $\approx 95k$ trajectories are long enough to carry a movable tail. The screen favors trajectories that already nearly serve the right areas, bending real behavior rather than inventing it. The screen only *nominates*: each nominee’s tail is then translated under the full objective. The pickup moves by the full offset δ , earlier tail states move by linearly tapered fractions of it, and the anchor state before the tail never moves. Edits physically route the trajectory through the under-served area before picking up without teleporting into it. The moved states carry supply with them, differentially, contributing presence to the same ΔS over the 5×5 neighborhood. Supply is thus *endogenous*: the optimizer sees the

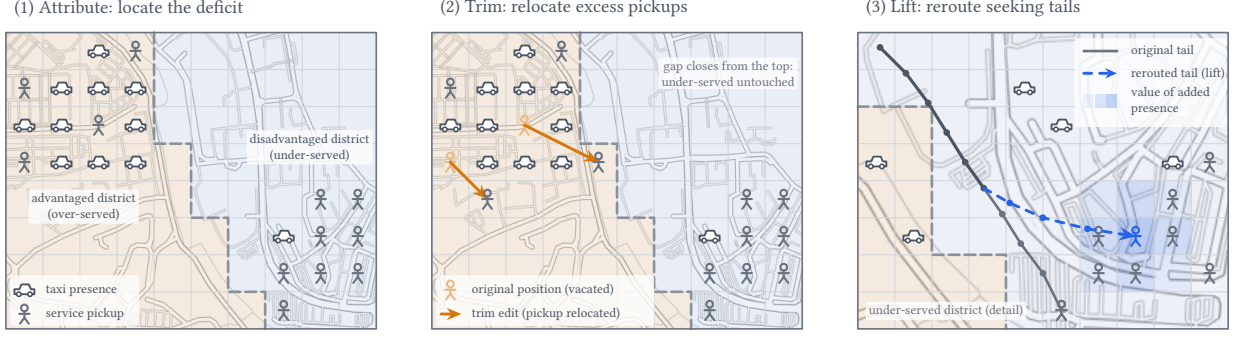


Figure 2: FATE on a stylized city: attribution, then the two editing phases. The street map is discretized into grid cells and split by a district boundary into an advantaged and a disadvantaged district; pickups and taxi presence are quantized to cells. (1) *Attribute*: taxi presence concentrates in the advantaged district while the disadvantaged district carries unmet pickups; attribution scores each trajectory’s contribution to the fairness deficit and selects the ones worth editing (§3.4). (2) *Trim* relocates excess pickups within the advantaged district, improving fairness metrics while the disadvantaged district gains nothing. (3) *Lift*, in a detail view of the under-served district: the rerouted tail bends the trajectory’s final seeking states into the cells where the supply gradient says added presence improves fairness most.

supply consequences of its own moves and is rewarded for providing service where it improves fairness. The fidelity term scores the actual rerouted tail at every iteration, and each completed edit updates a shared running state, so later edits optimize against the results of previous edits. Appendix B gives the taper weights, supply floor, and additional editor details.

Physical validity. Edits must satisfy the same king-move rule the source trajectories are filtered on (consecutive states move at most one cell per axis), enforced by an exact backward-reachability repair detailed in Appendix B. Infeasible edits are not applied: lift skips them in-loop, and the small fraction of trim edits with no compliant repair (about 5%) are reverted to the originals post hoc under the same rule, before computing fairness metric effects (§4).

Budget and phase order. Trim retains its selection by demand deficit attribution; lift fills the remaining budget with positive-score nominees (on the primary city, $k = 10,000$ splits into 2,455 trim and 7,545 lift edits), and the supply gradient is computed on the post-trim state. The two-phase structure is a scientific control, not an algorithmic necessity. Freezing trim keeps its optimization identical to the demand-only editor, which buys two properties: demand-only results reproduce exactly inside the combined run, and the trim-only versus trim+lift comparison is a clean ablation in which every difference is attributable to the new mechanism. A unified single-pass editor is a natural extension we leave as future work.

Downstream recipe: upweighted imitation. Editing alone is necessary but not sufficient. By design, the edited slice is roughly a tenth of the corpus, and a vanilla behavior-cloning objective that weights every demonstration equally averages it away and relearns the original bias; we verify this null empirically (§4). FATE therefore *upweights* the edited demonstrations in the imitation loss: instance reweighting for fairness [17], a member of the pre-processing family alongside disparate-impact repair [7], transplanted to imitation learning. Whether the recovery is specific to the *edit*, rather than an artifact of oversampling or of selecting fair data, is tested against two controls (upweighting a size-matched random subset

and upweighting the corpus’s fairest trajectories), with outcomes reported in §4. This intervention targets, on the training data, the same demonstration bias that in-processing regularization has addressed at the model level [37]; FATE moves the intervention to the demonstrations themselves.

4 EXPERIMENTS

4.1 Experimental Setup

Datasets. The primary city is Shenzhen: a 50-driver sample of real taxi GPS trajectories on a 48×90 grid of 0.01° cells, with district-level demographics. Raw data preprocessing included removing 106,677 pickups with almost no matching drop-offs — stuck GPS, not real demand. The external-validity city is San Francisco (Cabspotting): a 12-driver subsample matched on fleet density (sf12, 32×30 grid), with the same demographic feature names filled from ACS tract values. Density matching is required — the full 536-taxi fleet saturates the supply measure and leaves no editable fairness gradient.

Editor configuration. All edited corpora use the trim+lift editor of §3.4, with $\alpha = (0.1, 0.8, 0.1)$ weights selected using the three-class criterion of §4.5, $\epsilon = 2$, and the skip-on-infeasible rule of §3.4. Shenzhen: $k = 10,000$ (2,337 trim + 7,545 lift after filtering); San Francisco: $k = 2,000$ (1,330 trim + 629 lift).

Protocol and statistics. Every model-based comparison uses paired seeds (identical seeding before each arm), so per-seed differences cancel shared seed noise. With $n = 6$ paired seeds we read effects from the mean Δ , its t -interval, and dose-response monotonicity, and treat $p = 0.031$ only as a sign-unanimity certificate. Sign counts are written k/n : the delta was positive on k of the n seed pairs. The flagship w30 comparison is additionally run at $n = 12$ paired seeds in both cities, where an exact Wilcoxon p can survive multiple-comparison correction. External fairness metrics are compared before versus after editing on the data alone, with no model in the loop: one bootstrap resample of the active grid units is applied to both sides ($B = 1,000$, percentile), so the interval

Table 1: Shenzhen external fairness (migrant axis, district extremes; paired bootstrap): established instruments and group service levels, before → after editing ($B=1,000$).

metric	before	after	Δ [95% CI]
Disparate impact ($\uparrow$)	0.3325	0.3487	+0.0162 [+0.0136, +0.0189]
DP gap ($\downarrow$)	14.199	13.309	-0.890 [-0.992, -0.785]
Theil ($\downarrow$)	0.1550	0.1463	-0.0087 [-0.0097, -0.0076]
mean(Y disadv.) ($\uparrow$)	7.0734	7.1263	+0.0529 [+0.0086, +0.0989]
mean(Y adv.)	21.272	20.435	-0.837

isolates the edit. Units are spatially correlated and demographics are district-constant, so these intervals understate true uncertainty. Appendix D gives the exact-Wilcoxon floor arithmetic and why no driver-level bootstrap is available.

Metrics and claim discipline. We separate three metric classes and label every claim as follows: (i) *optimized* — F_{spatial} , F_{demo} , F_{fidelity} , the quantities the editor’s gradient sees; (ii) *design-targeted but not optimized* — the supply/demand ratios and mean(Y) per group that lift was aimed at, so improvement there is confirmatory; and (iii) *genuinely external* — demographic parity, disparate impact, the Theil index, and the distinct-taxi supply recount, absent from the objective. “Improves metrics it never optimizes” claims apply to class (iii) only. The external instruments are computed on the service ratio Y over active units, grouped per equity axis: *demographic parity* (DP; on a continuous outcome DP is the supply/demand-ratio gap — one number, reported once), *disparate impact* (DI), and the between-region *Theil* index, alongside the two group *levels* mean(Y | disadv.) and mean(Y | adv.), which are class (ii) context, not external instruments. The strict count of distinct external instruments is therefore {DI, DP/gap, Theil}. Grouping conventions and instrument definitions are in Appendix D.

4.2 Data-Level Fairness (Shenzhen)

Optimized metrics. Editing raises F_{demo} from 0.7988 to 0.8214 (+0.0226) and F_{spatial} from 0.1034 to 0.1095 (+0.0061) — class (i) numbers, reported for completeness rather than as evidence.

External metrics (class (iii)). Every established instrument moves toward fairness with bootstrap intervals excluding zero (Table 1). The gap closes from both ends: the advantaged group’s over-service *falls* while the disadvantaged group’s service *rises*.

The lift-up claim rides the supply channel. The mean(Y | disadv.) rise is a *design-targeted* (class (ii)) quantity — the lift mechanism was aimed at it — so we decompose it rather than celebrate it. Evaluating Y at $(S, D) \rightarrow (S, D') \rightarrow (S', D')$ splits the delta exactly into a demand move and a supply move ($B=2,000$). The demand channel (pickup relocation) carries large variance and is not significant, +0.0352 [−0.0095, +0.0814]. The supply channel (rerouted tails adding presence) is significant under *both* accounting conventions. Under the optimizer’s *fractional-presence* convention it is +0.0176 [+0.0128, +0.0225]; under the *distinct-taxi* accounting (supply re-counted from raw GPS; it reproduces the production supply grid exactly) it is +0.0411 [+0.0346, +0.0479].

Realism. Fidelity-A stability and lift’s expected Fidelity-B distributional cost (whole-tail relocation shifts trajectory-shape statistics more than a pickup-only move) are re-measured on the adopted corpus: Fidelity-A 0.844 vs. raw 0.848; corpus-level Fidelity-B (a

Jensen–Shannon divergence over aggregate mobility statistics [8]) at 0.187, concentrated in exactly the components a rerouted tail must move (Appendix C) — a by-design trade-off we disclose rather than net out.

Provenance disclosures. About 5% of trim edits (118/2,455) had no king-compliant repair and were reverted post hoc, before fairness metrics were computed. Before building lift we bounded the largest under-served service gain the supply channel could possibly deliver; that bound, and the far smaller realized gain, are in Appendix C.

Ablation: trim-only versus trim+lift. With trim frozen inside the combined editor (§3.4), the lift-disabled run reproduces the demand-only editor exactly, so every difference between the trim-only and trim+lift columns below isolates lift and certifies §3.3: demand-only editing improves the metrics without raising under-served service; the supply channel is what changes that.

Three metrics carry the qualitative story, and the structural expectation from §3.3 re-verifies exactly at the adopted weights: under trim-only, every external instrument still improves significantly (DP −0.611, DI +0.0098, Theil −0.0059, all CIs excluding zero) — but the disadvantaged group’s service level is unchanged while the advantaged level falls, so the improvement is pure top-down redistribution. Under trim+lift the disadvantaged level rises with a bootstrap interval excluding zero, and $\Delta F_{\text{spatial}}$ flips from negative to positive: trim only redistributes existing service; lift adds taxi presence where it was missing. On Shenzhen the trim-only versus trim+lift contrast is ΔF_{demo} +0.0146 versus +0.0226, $\Delta F_{\text{spatial}}$ −0.0011 versus +0.0061, and Δ mean(Y | disadv.) +0.000 (flat) versus +0.053 (CI excl. 0); on San Francisco it is ΔF_{demo} +0.0144 versus +0.0316 and migrant DP gap −0.052 versus −0.073 (both CIs excluding zero).

4.3 Downstream Propagation (Shenzhen)

Data quality (four sources). We score four corpora — raw, edited, BC-generated, and GAN-generated — on fairness and both fidelity axes, with generators trained from scratch for this comparison and Fidelity-A gated using the realism guardrail of §3.2; the claim under test is that the edited corpus is the fairest source that remains faithful on both realism axes. The claim holds: the edited corpus is the fairest source faithful to driver identity (Fidelity-A gate passed, matched 0.848 vs. mismatched 0.193), with Fidelity-B 0.187 the disclosed divergence of the bounded edits (Table 4, Appendix C). One instability we surface rather than smooth: the Shenzhen GAN’s Fidelity-B is bimodal across seeds, while the SF GAN stays uniformly healthy (§4.6); the per-seed spread is in Appendix C.

Vanilla training averages the edit away. A driver-conditioned BC policy trained on the edited corpus at uniform weight is not fairer than one trained on raw: paired ΔF_{demo} = +0.0016 ($n = 12$, 7/12 positive, $p = 0.11$, n.s.) — the null predicted in §3.4, and the reason the upweighting recipe exists.

Upweighting recovers it, edit-specifically. With the edited demonstrations upweighted, the paired gain is monotone in the weight and sign-unanimous: +0.0217 (w10) and +0.0267 (w20), each 6/6 (positive on all six seed pairs, the $p = .031$ floor). At the adopted dose we extend the flagship comparison to $n = 12$ paired seeds: +0.0297 ± 0.0029 (w30), 12/12 positive, exact Wilcoxon $p = .00049$,

t -interval $[+0.0278, +0.0315]$ — while both controls stay null at the same $n = 12$ (random -0.0017 , most-fair $+0.0012$, mixed signs, n.s.). Extending the dose saturates, with w30 at the knee, not a tuned endpoint (w40/w50 in Appendix C). Neither control reproduces it at any dose: the random-subset placebo is null-to-negative throughout, and the most-fair selection control reaches $+0.0033$ at w10 — significant, but under a sixth of the edited arm’s gain — then fades to n.s. at every higher dose. F_{spatial} also propagates on Shenzhen, though not on San Francisco (§4.6); per-dose values and the control contrast are in Appendix C. Fidelity-A is unchanged at every weight; Fidelity-B rises dose-monotonically — the expected signature of deliberately shifting population statistics toward the edited slice (Appendix C).

Model-level variance across paired retrainings is real but an order of magnitude below the upweighting effect (Appendix C).

4.4 Data-Augmentation Baselines

None of the baselines below optimizes a fairness objective; they are *editing-quality* baselines that test whether FATE’s gains come from its objective or merely from bounded perturbation (or corpus resampling) per se. Fairness is accordingly *expected not to improve* under the perturbation arms.

Perturbation arms. Three arms attack the same trajectories the headline edit selected, at the same $\varepsilon = 2$: iterative and single-step signed-gradient attacks on the frozen identity discriminator [15], and a seeded random-jitter placebo. The gradient arms are *iFGSM/FGSM with random restart* [11, 19] (PGD-style random initialization within the ε ball [23]); the random-start arms are the reported ones (the pre-registered $\delta = 0$ ablation: Appendix B). Results: Table 2.

One result lands against the pre-registered expectation: random jitter *does* raise F_{demo} , by $+0.0135$ — indiscriminately diffusing the selected trajectories diffuses their over-service too — but it does so by breaking the trajectories: 98.8% of its perturbed trajectories violate king-move adjacency and its distributional divergence reaches 0.447, against the edited corpus’s disclosed 0.187, while FATE edits are king-compliant by construction (per-arm figures in Appendix C). Identity fidelity survives every arm (≈ 0.846) — too small a perturbation to break driver identity, so adjacency and distributional axes carry the comparison. The three perturbation arms replicate on San Francisco — all well below the editor’s $+0.0316$, all violating adjacency wholesale — and the random-jitter fairness surprise does not recur there; random is the weakest mover.

Demographic oversampling. The naive lift-up baseline: duplicate real trajectories originating in demographically disadvantaged regions [17, 39] (all three equity axes, phantom driver IDs, rigid ± 1 -cell jitter), additively rebuilding both demand and supply at doses matched to FATE’s budget. At the matched duplication dose $d = 10,000$, targeted oversampling improves F_{demo} by $+0.0153$ (dose-monotone: $+0.0059, +0.0097, +0.0153$) — but it fabricates 10.5% of the corpus to get there, where FATE redistributes observed behavior at zero inflation and reaches $+0.0226$. The untargeted placebo (same fabrication without targeting) *degrades* fairness, -0.0172 , so the gain is targeting-specific; and the placebo’s DP gap explodes ($+2.8$) because fabricated supply lands mostly

Table 2: Cross-arm comparison at matched budgets (Shenzhen).

arm	ΔF_{demo}	$\Delta F_{\text{spatial}}$	inflation
raw (reference)	—	—	0%
FATE trim+lift	+0.0226	+0.0061	0%
iFGSM (rand. restart)	-0.0057	$+0.0126$	0%
FGSM (rand. restart)	$+0.0017$	$+0.0137$	0%
random jitter	$+0.0135$	$+0.0132$	0%
demog. oversampling (targeted)	$+0.0153$	$+0.0052$	10.5%
placebo (untargeted)	-0.0172	$+0.0138$	10.5%

in already-advantaged cells — a measured demonstration that ratio metrics can be gamed by fabrication, the demand-endogeneity concern of §3.3 made concrete. Two disclosures: the matched dose needs re-duplication (Appendix C), and fidelity is not scored — duplicates of real trajectories pass any realism check by construction.

Fairness-method baselines (training-side). The remaining comparison is against fairness methods proper, applied where they usually live: at training time, on the raw corpus, scored by the full paired protocol. Instance reweighing in the Kamiran–Calders style [17] — inverse group-service weights favoring disadvantaged-origin demonstrations — moves demand-adjusted fairness the *wrong* way (-0.0227 , all six seeds negative, $p = .031$) while paying the same distributional cost as w30 upweighting (Fidelity-B $+0.0144$), and its rolled-out policies still tilt allocation away from the disadvantaged group (-0.0010 , negative on all six, $p = .031$). An in-processing penalty in the spirit of [37] — the imitation loss plus λ times the demographic-parity gap in predicted next-cell mass — is inert at every trainable dose, destructive only where it dominates, identically under its signed and absolute-value formulations. The full λ grid, collapse magnitudes, and the 10^{-5} loss-share derivation are in Appendix C. Neither training-side intervention reproduces any part of the recovery that editing the demonstrations provides.

4.5 Robustness and Sensitivity

The control rows are re-measured deliberately: the random placebo is null-to-negative under every feature set, and the most-fair selection control — null on the primary set — is significantly *positive* under both alternate sets ($+0.0054$ and $+0.0072$ at w30, a fifth to a quarter of the edited arm’s gain). We report the strongest controls we found rather than average them away: even where selection captures a minority share, the edited arm’s gain is at least $3\times$ larger at every dose, which is the edit-specificity claim under its hardest test. Vanilla transfer likewise stays an order of magnitude below the upweighted gain under every set, reaching sign-unanimity only under the housing–GDP–compensation set ($+0.0022$, a tenth of its w30 gain).

Weight sensitivity. A weight sweep of six full editing runs, each scored on all three metric classes, shows the optimized ΔF_{demo} flat — within 0.001 — for $\alpha_{\text{sp}} \in [0, 0.55]$, while the class (ii) supply-channel lift-up declines monotonically with α_{sp} and is significant under both accounting conventions only for $\alpha_{\text{sp}} \leq 0.2$. The adopted (0.1, 0.8, 0.1) is the three-class criterion’s selection: any weighting in the flat range delivers the same ΔF_{demo} , and the low α_{sp} is what keeps the supply-channel lift-up significant. The full sweep and figure are deferred to Appendix C (Fig. 3).

Feature-set robustness. F_{demo} is specific to its demographic feature set, and its absolute magnitude shifts under each configuration, so every layer of the argument is re-measured under two alternate sets (Table 3, Appendix C). The design-targeted lift-up reproduces: the total $\Delta \text{mean}(Y \mid \text{disadv.})$ carries a bootstrap interval excluding zero on all three sets, and the supply channel is individually significant on all three under *both* accounting conventions (distinct-taxi: +0.0411, +0.0211, +0.0771).

Filtering is not a substitute. Deleting the K most-biased trajectories instead of editing them underperforms editing (0.7935 vs. 0.8214; Appendix C).

4.6 External Validity: San Francisco

The protocol reruns on San Francisco unchanged in algorithm, metric, and fidelity; because F_{demo} is city-specific and associational, San Francisco tests whether the *conclusions* reproduce.

Data level. F_{demo} rises 0.8752 $\rightarrow$ 0.9067 (+0.0316), F_{spatial} +0.0139, with Fidelity-A stability and lift’s Fidelity-B cost re-measured in the four-source comparison below (Appendix C). Every external instrument improves: Theil -0.0079 (CI $[-0.0094, -0.0065]$); the migrant-axis DP gap is significant under *both* grouping conventions, DI under one and n.s. under the other — a grouping dependence we state rather than hide (the SF sample is 12 drivers; per-convention values in Appendix C).

The supply channel replicates; the ratio tension is an accounting artifact. The supply channel is again positive and significant under *both* conventions: fractional-presence +0.0209 (CI $[+0.0122, +0.0300]$) and distinct-taxi +0.1027 (CI $[+0.0872, +0.1203]$) — the largest distinct-taxi supply effect of any corpus in this study (Shenzhen’s three feature sets: +0.0411, +0.0211, +0.0771; §4.5). But as on the prior corpus, lift also reroutes *pickups* into under-served cells, raising recorded demand there: the demand channel is significantly negative (-0.0533 , CI $[-0.0798, -0.0298]$), and under fractional-presence accounting the total mean($Y \mid \text{disadv.}$) moves net-negative (-0.0324 , CI excludes 0). We read this net-negative as a demand-endogeneity finding, not a failed replication: recorded demand in under-served areas was suppressed, and serving it raises the ratio’s denominator — the mechanism §3.3 predicts, observed in the wild. The distinct-taxi recount resolves the tension: the *total* turns significantly *positive* (+0.0493, CI $[+0.0185, +0.0790]$) — the real supply improvement outweighs the demand-side denominator increase. More rides *served* in under-served areas is a demand-side parity gain — DP, DI, and Theil all improve on San Francisco — while the supply channel independently replicates.

Downstream. The recovery reproduces on San Francisco. Vanilla BC is null (+0.0017, $n = 12$, 7/12 positive, n.s.); upweighting recovers fairness with a monotone dose-response: +0.0242 at w10 and +0.0309 at w20, each positive on all six seed pairs. At the adopted dose the flagship comparison reaches $n = 12$ here as well: +0.0333 $\pm$ 0.0050 (w30), 12/12 positive, exact Wilcoxon $p = .00049$, as on Shenzhen (§4.3), with both $n = 12$ controls far below the edited arm (control spreads, extended doses, and variance detail in Appendix C). One city difference stated plainly: F_{spatial} does *not* propagate on SF (-0.0040 at w30, n.s.; the primary-city result is in

§4.3). The four-source comparison also reproduces: the edited corpus is the fairest source while identity-faithful, with the full table in Appendix C.

San Francisco caveats. ACS foreign-born share proxies the migrant axis; the 12-driver density-matched subsample is small; and 14.9% of raw Cabspotting trajectories violate king-move adjacency at source (GPS gaps), so no corpus built from this data can be fully compliant. The standard we hold the editor to is therefore that its own edits introduce no violations (all of them comply), and the edited corpus ends *more* compliant than raw (87.6% vs. 85.2%). The Shenzhen GAN’s distributional collapse does not recur here (Fidelity-B 0.039, cf. §4.3); that cautionary note is city-specific.

5 CONCLUSION

FATE makes fairness a property of the demonstrations rather than of a constrained model or a synthetic corpus: it edits a small slice of real trajectories, chosen by attribution, under one bounded, realism-guarded objective, then upweights the edited demonstrations so the fairness survives imitation. On two cities the edits improve established fairness measures the objective never optimizes, and, unlike the demand-only editor whose gains never reached the under-served group, the supply channel adds statistically significant taxi presence to that group. Upweighted behavior cloning carries those gains into trained policies, and the control arms confirm the edit itself is what carries them.

The claims come with stated bounds. The optimized fairness term is associational: a partial R^2 over roughly ten district-level demographic profiles, with no identification and no individual-level reading. Adjusting for recorded demand treats demand as exogenous, but demand is suppressed exactly where service has been historically thin, so a demand-adjusted metric under-detects the inequity it audits — the same censoring that bounds any demand-side editor. With five or six paired seeds, no single test survives multiple-comparison correction; the one comparison extended to twelve paired seeds in each city — the flagship w30 recovery — does ($p = .00049$, 12/12 positive in both), and the remaining downstream evidence rests on effect direction and magnitude, confidence intervals, a monotone dose-response, and null controls rather than on any uncorrected p-value. San Francisco reproduces the Shenzhen conclusions with no algorithm change, but its baselines are city-specific and its magnitudes are not comparable. And the fidelity guardrail certifies that edited trajectories still read as their drivers — identity preservation, not fine-grained trajectory-shape realism.

Two directions follow naturally. The first is the single-pass editor §3.4 anticipates, in which supply is endogenous for every edit and the gradient decides each edit’s character. The second is transfer: whether other imitation objectives and adversarially trained generators inherit the edited corpus’s fairness as upweighted behavior cloning does, whether training-side allocation constraints can close the rollout allocation gap that data editing alone only attenuates (Appendix C), and whether the recipe holds on further cities.

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A DERIVATIONS

This appendix carries the closed-form details of F_{demo} deferred from §3.2. The projection form of Eq. (1) is what makes the measure usable as an *optimization objective* rather than a one-shot audit. Demographics are fixed attributes of the cells, so H is a constant of the optimization: as editing moves R , Eq. (1) implicitly re-fits the demographic regression exactly at every step, with no coefficients to refresh and nothing trained inside the editing loop. Nor is the $N \times N$ projection ever materialized: by the normal equations, $R^T(I - H)R = R^T R - (\tilde{X}^T R)^T (\tilde{X}^T \tilde{X})^{-1} (\tilde{X}^T R)$, so each evaluation costs a few length- N inner products. By the Frisch–Waugh–Lovell theorem [9, 20], the residualize-then-project construction recovers exactly the demographic coefficients of the full multiple regression, so r_{demo}^2 is interpretable as the share of demand-adjusted service inequity attributable to demographics; measuring fairness through the statistical dependence between protected attributes and outcomes follows the predictability logic of [7], with the direction reversed: there, disparate impact is certified by predicting the protected attribute from the data; here, the demographics predict the adjusted service residual.

Per-unit deficit partition. Because M and $I - H$ are idempotent, the explained share r_{demo}^2 of Eq. (1) admits the exact per-unit decomposition

$$r_{\text{demo}}^2 = \sum_i \frac{(MR)_i^2 - ((I - H)R)_i^2}{R^T MR}, \quad (4)$$

a mathematically exact partition of the fairness deficit across active units; a signed variant separates over- from under-served units. This is the demand-deficit attribution that drives trim (§3.4).

Supply-gradient closed form. The F_{demo} component of the value-of-presence map of §3.4 has the closed form

$$\frac{\partial F_{\text{demo}}}{\partial S_i} = \frac{2}{R^T MR} \frac{((I - H)R)_i - F_{\text{demo}}(MR)_i}{\max(D_i, d_0)}, \quad (5)$$

against which the automatic gradient is verified.

Spatial-equity closed form. The F_{spatial} term of §3.2 scores the inequality of two supply-normalized service rates — the demand-service ratio $\text{DSR}_i = D_i/S_i$ and the arrival-service ratio $\text{ASR}_i = A_i/S_i$, with A_i the unit's recorded dropoffs and D_i, S_i as in §3.1 — with the Gini index, an inequality measure widely used in transportation equity research [14, 18]. The Gini enters as a pairwise mean absolute difference, differentiable almost everywhere and hence usable inside the gradient objective:

$$\begin{aligned} \text{Gini}(x) &= \frac{\sum_i \sum_j |x_i - x_j|}{2N^2 \bar{x}}, \\ F_{\text{spatial}} &= 1 - \frac{1}{2} (\text{Gini}(\text{DSR}) + \text{Gini}(\text{ASR})). \end{aligned} \quad (6)$$

B EDITOR IMPLEMENTATION DETAILS

This appendix collects the editor implementation details deferred from §3.4.

Soft cell assignment. The discrete grid is bridged by temperature-annealed soft cell assignment [3, 16, 22] over the same 5×5 window used to aggregate supply, moving from broad early exploration to precise late assignment; the best iterate is kept.

Editor constants. In lift, the pickup moves by the full offset δ and earlier tail states move by linearly tapered fractions $w_j = j/\ell$ ($0.25/0.5/0.75/1.0$ at $\ell = 4$), with the anchor state before the tail fixed. Each moved seeking state contributes $1/12$ of an hourly presence unit to each cell of its 5×5 neighborhood in the supply perturbation ΔS , and the objective evaluates supply as $\text{clamp}(S + \Delta S, s_0)$ with supply floor $s_0 = 0.1$.

King-move repair. The source data is preprocessed — following the pipeline of [36] — to satisfy a king-move rule: consecutive states move at most one cell per axis, $\max(|dx|, |dy|) \leq 1$. Edits should not be held to a lower standard than the data they edit; a bounded two-cell pickup move would silently violate the very rule the source trajectories are filtered on. Discretized edits therefore pass through an exact backward-reachability repair that returns the king-compliant assignment nearest the tapered targets, or reports that none exists.

Linearized-offset screen. The candidate screen of §3.4, in detail. Each trajectory's tail — its last ℓ seeking states and pickup — is rigidly translated by every nonzero integer offset $\delta \in [-\varepsilon, \varepsilon]^2$, and each translation is scored by its linearized gain: the value-of-presence collected at the tail's new positions minus that at its old ones, each state weighted by the presence mass it carries and the

Table 3: Feature-set robustness under trim+lift (Shenzhen; Δ rows are after – before edit). *-marked rows: every cell's paired-bootstrap 95% interval excludes zero ($B = 2,000$ for $\Delta \text{mean}(Y \mid \text{disadv.})$; $B = 1,000$ for DI/DP/Theil).

	{hous,comp,migr}	{hous,gdp,comp}	+logpopdens
before-edit F_{demo}	0.7988	0.8069	0.7253
editor ΔF_{demo}	+0.0226	+0.0206	+0.0220
$\Delta \text{mean}(Y \mid \text{disadv.})^*$	+0.0529	+0.0594	+0.1461
DI Δ (migrant/extremes)*	+0.0162	+0.0147	+0.0191
DP gap Δ^*	-0.890	-0.787	-0.886
Theil Δ^*	-0.0087	-0.0080	-0.0085
L1: edited fairest faithful?	see Tab. 4	yes	yes
vanilla-BC transfer (w1)	see §4.3	+0.0022	+0.0011 (n.s.)
weighted-BC w30	see §4.3	+0.0248	+0.0256
random placebo (w30)	-0.0023 (n.s.)	-0.0019 (n.s.)	-0.0044 (n.s.)
most-fair select (w30)	+0.0009 (n.s.)	+0.0054	+0.0072
variance (b0 vs FATE)	see §4.3	null	null

map summed over the same 5×5 window through which presence spreads (§3.1). A trajectory's score is its best offset.

Perturbation-arm stationarity. Detail for the baselines of §4.4. We pre-registered $\delta = 0$ -initialized ablation arms expecting a no-op — a difference-mode Siamese head has zero subgradient at an identical pair — but the deployed discriminator compares embeddings by *concatenation*, which is not stationary there: measured, the $\delta = 0$ arms attack rather than stall (the iterative arm lands within 0.0003 of its random-start twin; the single-step arm differs with initialization, as a one-step method must). The stationarity assumption held only for the head we did not deploy.

C EXTENDED RESULTS

Weight sensitivity. Fig. 3 reports the full weight sweep summarized in §4.5: six full editing runs across the simplex (fidelity fixed at 0.1), each scored on all three metric classes. The optimized ΔF_{demo} is flat — within 0.001 — for $\alpha_{\text{sp}} \in [0, 0.55]$; the class (ii) supply-channel lift-up declines monotonically with α_{sp} and is significant under both accounting conventions only for $\alpha_{\text{sp}} \leq 0.2$; and class (iii) instruments improve significantly at every point. The adopted (0.1, 0.8, 0.1) is the three-class criterion's selection: any weighting in the flat range delivers the same ΔF_{demo} , and the low α_{sp} is what keeps the supply-channel lift-up significant.

Feature-set robustness. Extending the summary in §4.5: the channel mix behind the reproduced lift-up is not constant — on the primary set the demand channel is n.s., while under both alternate sets it is significant and carries most of the larger totals — so the added-presence mechanism replicates while the demand-side share varies with the feature set, and we report the decomposition rather than the total alone. Table 3 gives every layer of the argument under the two alternate sets.

Filtering versus editing. Detail for the filtering rebuttal of §4.5. Deleting the K most-biased trajectories rather than editing them inverts the gain: F_{demo} falls to 0.7935 at the maximum $K = 2,455$, where editing the same attribution-selected slice raises it to 0.8214 — so neither removing nor selecting data (§4.3) substitutes for editing it.

Supply-channel upper bound. Supporting the provenance disclosure of §4.2. Before the lift phase was built, an unconstrained greedy search bounded the largest achievable gain at +0.786 on

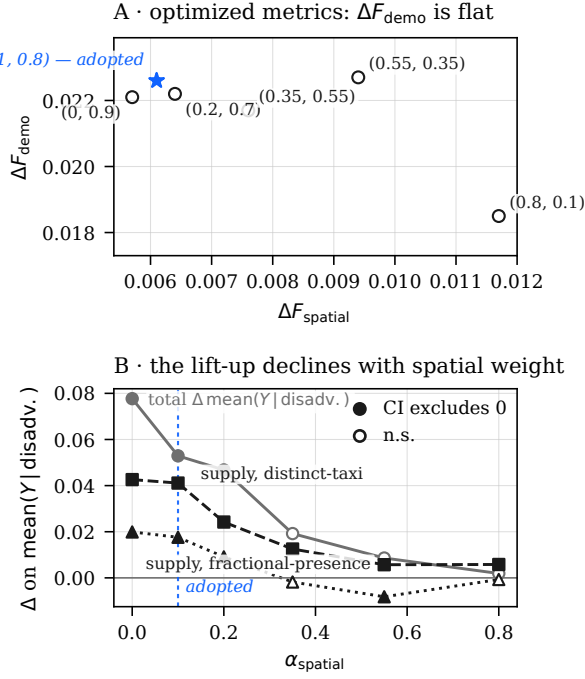


Figure 3: The weight sweep, scored beyond the optimized metrics. (A) On the optimized plane, ΔF_{demo} is flat across $\alpha_{\text{sp}} \in [0, 0.55]$. (B) The design-targeted lift-up declines monotonically with spatial weight and is significant under both accounting conventions only for $\alpha_{\text{sp}} \leq 0.2$ (filled = 95% CI excludes 0). The adopted (0.1, 0.8, 0.1) is the three-class criterion’s selection.

$\text{mean}(Y | \text{disadv.}) - 2.6 \times$ the pre-registered threshold of +0.3 for proceeding with the build (full accounting +0.882); the realized +0.053 sits far below this realism-free ceiling, as a realism-bounded editor should.

The two realism axes. *Fidelity-A* is the identity axis: the frozen driver-identity discriminator’s mean score on matched pairs (an edited trajectory against a trace of its original driver; higher is better), the evaluation-time reading of the §3.2 guardrail, and validated by its separation from mismatched-driver pairs. *Fidelity-B* is the distributional axis: a Jensen–Shannon divergence over aggregate mobility statistics, computed against the raw corpus (raw scores 0); lower is better.

Edited-corpus Fidelity-B components. Supporting the realism disclosure of §4.2. The corpus-level Fidelity-B of 0.187 concentrates in exactly the components a rerouted tail must move — terminal cell (0.635), net displacement (0.147), coverage (0.142) — while trajectory length, which neither editing phase alters, barely registers (0.060).

Four-source data-quality comparison (Shenzhen). The full table behind the four-source claim of §4.3: raw, edited, BC-generated, and GAN-generated corpora scored on fairness and both fidelity axes (means over five seeds; the GAN row’s Fidelity-B is seed-bimodal, detailed next).

Table 4: Data-quality comparison, four sources (Shenzhen; means over five seeds; the GAN row’s Fidelity-B is seed-bimodal — see text).

source	F_{demo}	Fid.-A ($\uparrow$)	Fid.-B ($\downarrow$)
raw	0.7988	0.848	0.000
edited (trim+lift)	0.8214	0.844	0.187
bc-generated	0.7980	0.848	0.011
gan-generated	0.8089	0.848	0.171

Shenzhen GAN per-seed spread. Supporting the four-source disclosure of §4.3. The Shenzhen GAN’s Fidelity-B mean of 0.171 hides a bimodal split across seeds (± 0.129 ; three of five seeds at 0.197–0.295, two at 0.03–0.04), and its worst seeds fail broadband — length-component divergence 0.247 where the edited corpus sits at 0.060.

Upweighting dose saturation (Shenzhen). Supporting §4.3. Beyond the adopted w30, the edited arm’s gain extends to +0.0323 (w40) and +0.0339 (w50), each 6/6, with the per-step increment shrinking monotonically (+0.0050, +0.0035, +0.0021, +0.0016) — w30 sits at the knee.

Downstream F_{spatial} and fidelity dose detail (Shenzhen). Supporting §4.3. F_{spatial} propagates from +0.0008 at uniform weight to +0.0052 at w30 (each $p = .031$), while at w30 both controls significantly degrade it. Fidelity-A is unchanged at every weight (all n.s., $|\Delta|$ at most +0.0007); Fidelity-B rises dose-monotonically from 0.020 to 0.027 absolute, against raw 0.013.

Oversampling re-duplication (Shenzhen). Supporting §4.4. Only 8,241 distinct disadvantaged-origin trajectories exist, so meeting the matched dose $d = 10,000$ requires 1,759 draws that repeat earlier ones.

Model-level variance ($n = 5$ versus $n = 10$). Supporting §4.3. A paired baseline-vs-FATE comparison at uniform weight gives $\Delta F_{\text{demo}} = +0.0030 \pm 0.0022$ at $n = 10$ seeds (9/10 positive; Wilcoxon $p = .0039$): the training procedure alone transmits a small but real fraction of the edit, an order of magnitude below the upweighted gains, so the weighting lever remains what carries the transfer. This $n = 10$ effect size is unchanged from the original five-seed suite (+0.0031) and now clears conventional significance outright (t -test $p = .0019$).

Fairness-penalty λ grid. Detail for the in-processing penalty baseline of §4.4. The penalty was tested in both a signed formulation (advantaged minus disadvantaged, the direction the data over-erves) and an absolute-value formulation. It is inert at every dose that trains stably (six-seed suites: $|\Delta F_{\text{demo}}| \leq 0.0009$ signed at $\lambda \in \{1, 3, 16, 10\}$, +0.0008 absolute at $\lambda = 10$, all n.s.; single-seed probes at $\lambda = 100$ agree; the term contributes on the order of 10^{-5} of the loss), and destructive only at the dose where it finally dominates: at $\lambda = 1000$ the signed form collapses F_{demo} by -0.2053 with Fidelity-B at 0.514, the absolute form by -0.1293 (negative on all six seeds) with Fidelity-B at 0.363, against raw’s ≈ 0.013 . The absolute formulation’s shallower collapse is the gap pushing back once it crosses zero; the two formulations track each other at every λ .

Perturbation-arm per-arm figures. Supporting §4.4. On San Francisco the three perturbation arms violate king-move adjacency

wholesale (72.0/87.0/97.5%) and pay 2–4× the edited corpus’s distributional divergence (0.198–0.418 against 0.098). On Shenzhen the gradient arms violate king-move adjacency (54.4% iFGSM, 91.4% FGSM) with distributional divergence 0.143 and 0.339 respectively, against the edited corpus’s 0.187.

San Francisco downstream detail. Supporting §4.6. At $n = 12$ the controls sharpen the contrast rather than soften it: the random placebo turns significantly *negative* (-0.0077 , positive on only 2 of 12, $p = .0024$), and the most-fair selection control is significantly positive but about a sixth of the edited arm’s gain ($+0.0059$, 10 of 12 positive, $p = .0020$; at $n = 6$ per dose it was significantly negative at w_{10} and n.s. at w_{20}). Extended doses: $+0.0370$ (w_{40}) and $+0.0371$ (w_{50}), each 6/6, flat past w_{40} ; the random control stays null-to-negative and the most-fair selection control stays significantly *positive* ($+0.0063/+0.0066$) — the dominated-control pattern of the alternate feature sets (§4.5). External-instrument grouping detail: the DP gap is -0.0729 [$-0.1132, -0.0259$] under district extremes and -0.0370 [$-0.0733, -0.0002$] under median split; DI is $+0.0058$ [$+0.0004, +0.0103$] under district extremes and n.s. under median split. at $n = 10$ seeds ($\Delta F_{\text{demo}} = -0.0009 \pm 0.0035$, mixed signs, closer to zero than at $n = 5$), unlike the primary city’s small significant positive effect. Four sources (generators retrained from scratch; identity gate passed, matched 0.958 vs. mismatched 0.034): the edited corpus is the fairest source (F_{demo} 0.9067 vs. GAN 0.8795, raw 0.8752, BC 0.8744) while identity-faithful (Fidelity-A 0.9581 $\approx$ raw 0.9578; Fidelity-B 0.098, the expected bounded-edit divergence).

Rollout allocation shares (both cities). Supporting the future-work discussion of §5. Beyond metric propagation, we rolled out the trained policies and measured the share of pickups they allocate to disadvantaged areas. That share still tilts slightly negative: on Shenzhen, -0.0049 at w_{30} on the demand-only corpus and -0.0033 under trim+lift (negative on all six seed pairs in both suites, $p = .031$), attenuated by roughly a third, not reversed; San Francisco replicates the pattern at roughly ten times the magnitude (-0.0432 demand-only, -0.0335 trim+lift, again negative on all six seed pairs in both suites), roughly 22% attenuation. Seeking-state shares are n.s. at every weight in both suites. The paper claims the data-level lift-up and the metric propagation; this allocation gap is what motivates training-side allocation constraints as future work.

D STATISTICAL PROTOCOL DETAILS

This appendix collects the statistical-protocol details deferred from §4.1.

Exact-Wilcoxon floor arithmetic. With $n = 6$ paired seeds the two-sided Wilcoxon signed-rank p floors at 0.03125, attained exactly when all six seeds share a sign, which is why $p = 0.031$ reads only as a sign-unanimity certificate. The flagship w_{30} comparison at $n = 12$ paired seeds has floor .00049, low enough for an exact Wilcoxon p to survive multiple-comparison correction. Suites with $n = 5$ seeds cannot reach $p < 0.05$ (floor 0.0625) and are reported as effect-versus-noise.

Why the bootstrap resamples units. A driver-level bootstrap is unavailable rather than declined: the demand grid is a mean-hourly artifact, not a per-driver sum, and supply is environmental, so Y does not decompose into driver contributions. Active grid units are the only available resampling level.

Instrument definitions. The service ratio Y is grouped per equity axis (housing price, compensation, migrant share) under two conventions, district extremes and median split. Demographic parity (DP) is the signed gap between the group levels $\text{mean}(Y \mid \text{disadv.})$ and $\text{mean}(Y \mid \text{adv.})$; disparate impact (DI) is their ratio ($1 = \text{parity}$); the Theil index is between-region and grouping independent.